

Unit 1: Introduction to Functions

1. Classify the function family for each equation or graph.

$y = (x + 2)^3$	Function Family: <u>Cubic</u>
$y = x + 23$	Function Family: <u>linear</u>
$y = -2^{(x-7)}$	Function Family: <u>exponential</u>
$y = (x)^2 + 22$	Function Family: <u>quadratic</u>

2. Find each value given the function $f(x) = \frac{x}{3} + 6$

a. $f(-12) = ?$

$$\begin{aligned} & \frac{-12}{3} + 6 \\ & -4 + 6 \\ & \textcircled{2} \end{aligned}$$

b. Find the input for which $f(x) = 21$

$$\begin{aligned} 21 &= \frac{x}{3} + 6 \\ -6 & \quad -6 \\ \hline 3(15) &= \left(\frac{x}{3}\right) \cdot 3 \\ 45 &= x \end{aligned}$$

3. Rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$.

a. $(\sqrt[5]{82})^3$

$$82^{\frac{3}{5}}$$

b. $(\sqrt{6^3})$

$$6^{\frac{3}{2}}$$

c. $\frac{1}{(\sqrt[6]{24})^3}$

$$24^{-\frac{1}{2}}$$

d. $\frac{1}{\sqrt[3]{8}}$

$$8^{-\frac{1}{3}}$$

4. Write an equivalent expression for each radical expression.

a. $\sqrt{32}$

$$\begin{aligned} & \sqrt{32} \\ & \begin{array}{c} \swarrow \searrow \\ 8 \quad 4 \\ \swarrow \searrow \quad \swarrow \searrow \\ 4 \quad 2 \quad 2 \quad 2 \\ \textcircled{2} \quad 2 \quad 4\sqrt{2} \end{array} \end{aligned}$$

b. $\sqrt{252}$

$$\begin{aligned} & \sqrt{252} \\ & \begin{array}{c} \swarrow \searrow \\ 42 \quad 6 \\ \swarrow \searrow \quad \swarrow \searrow \\ 7 \quad 6 \quad 3 \quad 2 \\ \textcircled{3} \quad 2 \quad 6\sqrt{7} \end{array} \end{aligned}$$

c. $\frac{\sqrt[3]{384}}{\sqrt[3]{81}}$

$$\begin{aligned} & \frac{\sqrt[3]{384}}{\sqrt[3]{81}} \\ & \frac{4\sqrt[3]{2}}{3} \end{aligned}$$

Handwritten prime factorizations for 384 and 81 are shown to the right of the fraction.

5. Multiple or divide the following radicals.

a. $\frac{6}{\sqrt{7}}$

$$\frac{6 \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$\frac{6\sqrt{7}}{7}$$

b. $\frac{\sqrt{54x^2}}{\sqrt{3x}}$

$$\frac{3x\sqrt{6}}{\sqrt{3x} \cdot \sqrt{3x}}$$

Handwritten prime factorizations for 54 and 3 are shown to the right of the fraction.

$$\frac{3\sqrt{18x}}{3x}$$

Handwritten prime factorization for 18 is shown to the right of the fraction.

$$\frac{6x\sqrt{18x}}{3x} \rightarrow 3\sqrt{2x}$$

c. $2\sqrt{3xz} \cdot \sqrt{6xy^5}$

$$\begin{aligned} & 2\sqrt{18x^2y^5z} \\ & 6xy^2\sqrt{2yz} \end{aligned}$$

Handwritten prime factorizations for 18 and 6 are shown to the right of the expressions.